

Phishing Detection — Deliverable 1 Report:

Github Repository: <https://github.com/FahadAftab14/Phishing-Detection>

1. Original Dataset Fields (Before Feature Engineering)

Four raw datasets were used. Fields available before any bucket-based processing:

- Nazario (email_text_cleaned.csv): label, text, message_id, date, from, to, subject, body, content_type, parse info.
- Enron (enron_emails_cleaned_temporal.csv): message, message_id, date, from, to, cc, bcc, subject, body, plus derived UTC/local time fields and temporal anomaly flags.
- Nazario (nazario5_cleaned_temporal.csv): sender, receiver, date, subject, body, urls, url_count, plus same derived temporal fields as Enron.
- PhishTank (PhishTank_2026_processed.csv): URL, label, domain, is_ip_based, subdomain_count, path_length, query_param_count, url_length, uses_https.

No dataset contained sender IP, email authentication results (DKIM/SPF/DMARC), or domain age/reputation data.

2. Bucket A — Real, Readily Available Data

Built directly from existing fields, no data generation involved.

- Sender identity and domain — pulled from existing from/sender fields.
- Sender volume and activity window — email count, active days, average daily volume per sender.
- Timing patterns — day of week, hour of day, weekend flag (already present in source data).
- Inter-arrival gap between a sender's own emails — used as a stand-in for arrival delay, since real transit-delay data does not exist in any dataset (no Received headers survived in Enron; Nazario had only 3/200 incidental cases, phishing-only, so excluded).
- URL count per email — existing field for Nazario, regex-extracted for Enron.

3. Bucket C — Synthetic Data

Why synthetic data was needed

The behavioral layer of the model requires sender IP reputation, domain reputation/age, and DKIM/SPF/DMARC results. This system is meant to run in real time, but none of the four public datasets contain this information, and it cannot be derived from what they do contain. This data had to be generated instead, using real published statistics as a basis rather than deriving it from the label directly.

Sources used to model probability distributions

- DMARC.org — global DMARC adoption stats — <https://dmarc.org/stats/dmarc/>
- Valimail — DMARC/DKIM/SPF adoption explainer: <https://www.valimail.com/blog/dmarc-dkim-spf-explained/>
- Validity — DMARC adoption deep dive: <https://www.validity.com/blog/dmarc-adoption-a-deep-dive-into-the-current-state-of-email-authentication/>
- Interisle / Cybercrime Information Center — domain age at time of phishing report: <https://interisle.substack.com/p/do-criminals-age-domains>
- Interisle — Phishing Landscape 2025 (TLD abuse scores): <https://www.cybercrimeinfocenter.org/terminology>
- IPinfo x AbuseIPDB — VPN/residential proxy abuse study, RSAC 2026: <https://ipinfo.io/blog/vpns-residential-proxies-abuse-findings-rsac-2026>
- ARIN — DNS Research Federation grant report: <https://www.arin.net/blog/2026/01/08/2024-grant-report-dnsrf/>

Conclusions drawn from these sources

- DMARC: about 50% of domains globally have a DMARC record; of those, about 44% are enforced (quarantine/reject). Large corporate domains adopt at a much higher rate (~90%+ record, ~44% enforced).
- Domain age: real phishing domains are mostly fresh — 23% under 30 days old, 81% under 1 year old. The remaining 19% are aged domains, so age alone is not a fully reliable signal.
- TLD risk: cheap/rare TLDs are far more abused than common ones (e.g. .xin scored 10,810 vs .com at 30 on Interisle's phishing scale).
- IP reputation: about 45–53% of abusive IPs are linked to residential proxies or VPNs, versus corporate mail traffic which is overwhelmingly datacenter-origin.

What was generated

- Domain age and reputation score — sampled per domain using the age distributions above, combined with TLD risk weighting and random noise.
- DKIM / SPF / DMARC pass, fail, and enforcement flags — sampled per domain using the adoption rates above, correlated with each other rather than independent.
- Sender IP behavior and reputation score — residential proxy / VPN / datacenter flags sampled per email using the rates above.
- URL structural features — real PhishTank rows sampled (with replacement) onto phishing emails containing a URL; synthetic, assumption-based values used for benign emails containing a URL, since PhishTank has no benign examples.

All synthetic values were generated per class (benign vs phishing) using the probabilities above, not copied from the label directly, and checked afterward to confirm no single synthetic feature separates the classes perfectly.

4. Final Dataset Overview

- Output file: features_step5_full_synthetic.csv — 253,229 rows, 32 columns.
- Combines Enron (benign) and Nazario (phishing) at the email level.
- Contains real contextual/temporal fields (Bucket A) alongside synthetic behavioral and URL fields (Bucket C).
- Class split: 250,249 benign vs 2,980 phishing — imbalance to be handled at model-training stage.
- PhishTank retained separately as the real-URL source table used to populate phishing-side URL features.
- Known open item: synthetic benign URL parameters need retuning — current values do not clearly separate from real phishing URL structure and should be adjusted before model training.